formulated initially by Paul Samuelson (1969), who won the Nobel Prize in 1970, and Robert Merton (1969, 1971). Merton won the Nobel Prize in 1997 with Myron Scholes, one of the creators of the Black–Scholes (1973) option pricing model, for the valuation of derivatives. As we'll see shortly, the solution to the dynamic portfolio choice problem is intimately related to derivative valuation, and both employ the same economic concepts and solution techniques.

2. The Dynamic Portfolio Choice Problem

An investor facing a dynamic portfolio choice problem has a long horizon, say ten years, and can change her portfolio weights every period. A period could be one year, which is common for retail investors meeting with their financial planners for an annual tune up, or one quarter, which is common for many institutional investors. For high-frequency traders, a period could even be even shorter than one minute. The portfolio weights can change each period in response to time-varying investment opportunities as the investor passes through economic recessions or expansions, in response to the horizon approaching (as she approaches retirement, say), and potentially in response to how her liabilities, income, and risk aversion change over time. In this section, we abstract from the last of these considerations and assume that she has no liabilities and no income and is (fortuitously) given a pile of money to invest. (We introduce liabilities in section 3 and consider income in chapter 5.) We also assume her risk aversion and utility function remain constant.

2.1. DYNAMIC TRADING STRATEGIES

At the beginning of each period t, the investor chooses a set of portfolio weights, x_t. Asset returns are realized at the end of the period t+1, and the portfolio weights chosen at time t, x_t, with the realized asset returns lead to the investor's wealth at the end of the period, W_{t+1}. The wealth dynamics follow

$$W_{t+1} = W_t(1 + r_{p,t+1}(x_t)), \tag{4.1}$$

where wealth at the beginning of the period, W_t, is increased or decreased by the portfolio return from t to t+1, $r_{p,t+1}(x_t)$ and this is a function of the asset weights chosen at the beginning of the period, x_t.

I illustrate this in Figure 4.1 for a dynamic horizon problem over T=5 periods. At the beginning of each period the investor chooses portfolio weights, x_t. These weights, together with realized asset returns, produce her end of period wealth, W_{t+1}, following equation (4.1). The procedure is repeated every period. The sequence of weights over time, $\{x_t\}$, is called a *dynamic trading strategy*. It can